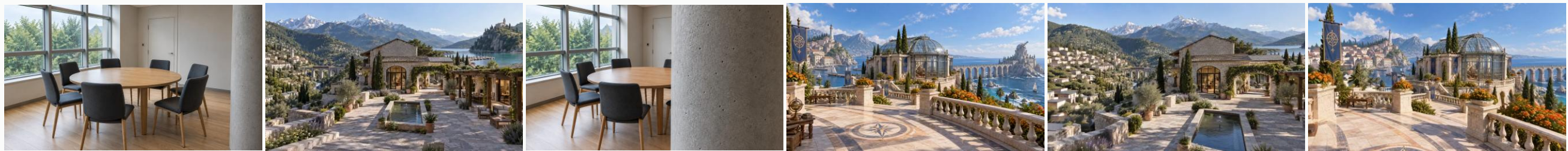


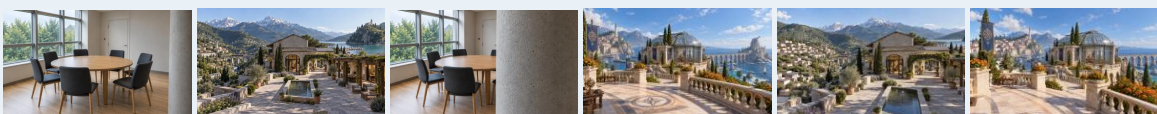
Spatial Memory Management

Disorganized and redundant spatial memory



Existing Methods

Compression-based methods



Compression



Key spatial evidence may be lost

Retrieval-based methods

Turn right ➡



Current state

Retrieve

Geometry-based



Occluded view

Semantic-based



Semantically similar, wrong layout

Without spatial reasoning, retrieval can fail in multiple ways

SMI: Understanding-driven method

Spatial organization



Scene A

Scene B

Scene C

Understanding-driven retrieval



Current view

Turn right ➡



Right-turn candidate

The Benefits of SMI

Memory sparsification

Base 0%

SMI

83.68%



Improved spatial consistency

0 s

16 s

37 s

Base



SMI



Improved generation stability

0 s

45 s

60 s

Base



SMI

